

Credit Card Approval Prediction System

Project Description:

The Credit Card Approval Prediction application utilizes data analytics and automated decision logic to streamline the credit underwriting process. It offers financial institutions and users a fast, data-driven experience to assess application risks instantly. The platform includes an interactive frontend dashboard that takes key applicant parameters (such as income, credit score, and employment history), processes them through a backend validation engine via a custom API endpoint (/predict), and displays immediate, color-coded risk classifications.

Built with the Flask web framework, the system runs locally as a development server to ensure responsive, low-latency processing. With secure parameter management and robust input validation, the application empowers credit teams to evaluate consumer risk profiles cleanly and with high structural confidence.

Scenarios

Scenario 1:

A user enters a profile with a high credit score, stable income, and low existing debt. The system processes the input via the /predict route and instantly displays a green approval confirmation banner alongside a calculated low-risk index score.

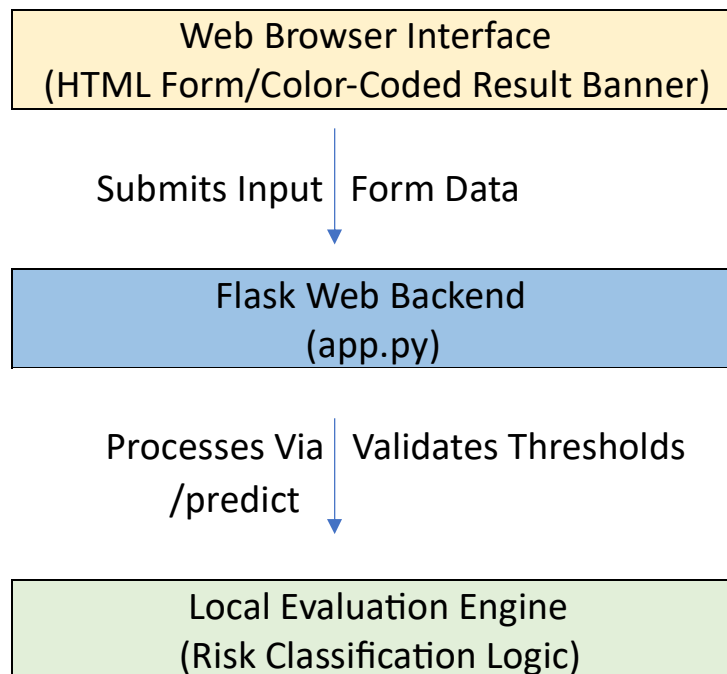
Scenario 2:

A user submits an application with an ultra-low credit score and multiple recent payment defaults. The application flags the profile, routes it through the high-risk verification logic, and returns a clear red rejection indicator to prevent subprime exposure.

Scenario 3:

A data analyst runs a performance validation test by passing edge-case parameters (like zero income or missing fields) into the form. The system handles the inputs gracefully, returning a structured validation error message without crashing the backend web server.

Technical Architecture:



Description:

The Credit Card Approval Prediction system implements a clean two-tier client-server architecture to deliver instant risk assessment. The user interface provides a responsive form built with

HTML and CSS, while the Flask-based backend orchestrates data validation, condition evaluation, and route handling. It interfaces directly with local rule parameters to generate tailored application results safely and securely on a local development server.

Pre-requisites:

Flask Framework Knowledge: [Flask Documentation](#)

HTML, CSS, and Form Handling: [W3Schools HTML/CSS Tutorials](#)

Python Programming Proficiency: [Python Documentation](#)

Environment Configuration Setup: [Python-dotenv](#) Guide